Supplementary Material

Standardised Comparison of Convolutional and Transformer Backbones for Brain Tumour MRI Classification under Perceptual-Hash Deduplication

Parth Mital, Ved Lawand, Biji C. L., Muzaffar Ahamad Dar, George Priya Doss C.

This document accompanies the main manuscript. Item numbers are those cited in the main text. Unless stated otherwise, every quantity is measured on the held-out test split, which was evaluated once per model using the checkpoint selected on the validation split. The per-model artefacts from which every table and figure here was produced are released with the code repository described in the main manuscript.

S1. Dataset characterisation

Figures S1 and S2 show the native image dimension and intensity distributions of the two experimental tracks, summarised numerically in Table 3 of the main text. Figure S3 shows the class distribution of the binary detection dataset, and Figure S4 the stratified training, validation, and test split distributions of both tracks, whose per-class counts are given in Table 4 of the main text.

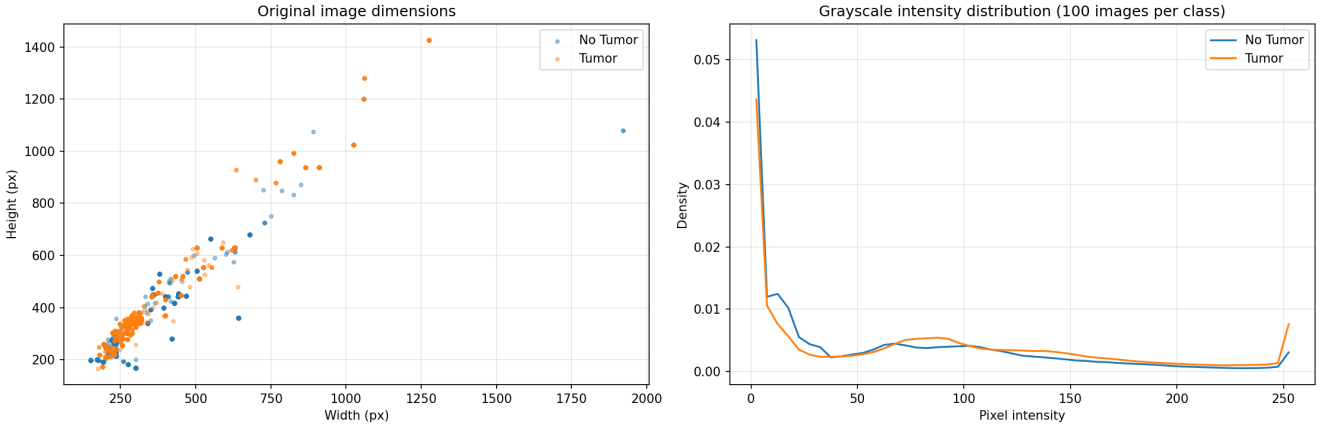


Figure S1: Native image dimension and intensity distributions for the binary detection dataset, showing the wide variation in resolution and field of view across the two contributing sources.

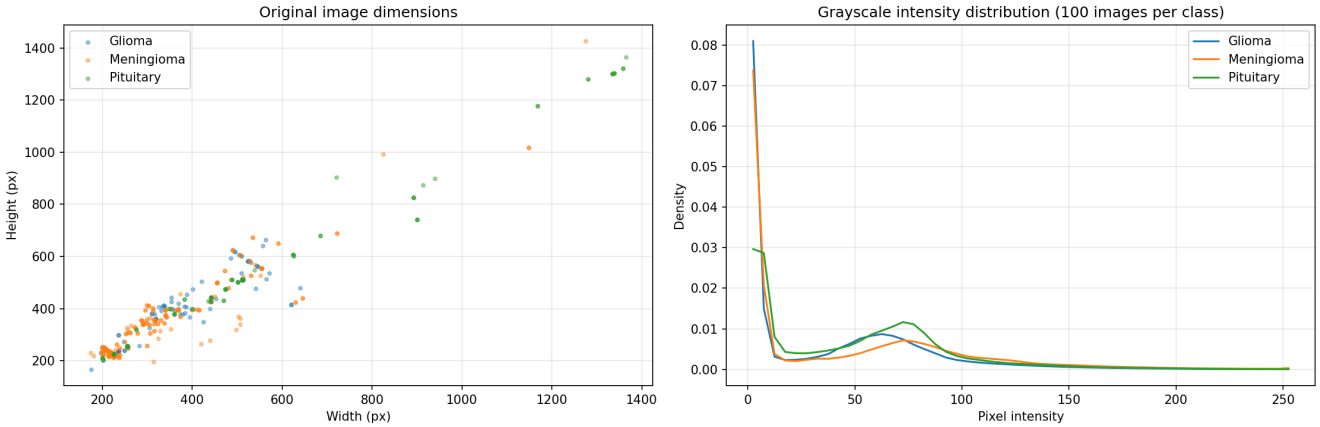


Figure S2: Native image dimension and intensity distributions for the multiclass classification dataset before deduplication, which is markedly more homogeneous than the binary collection, with a median resolution of 512×512 pixels in every class.

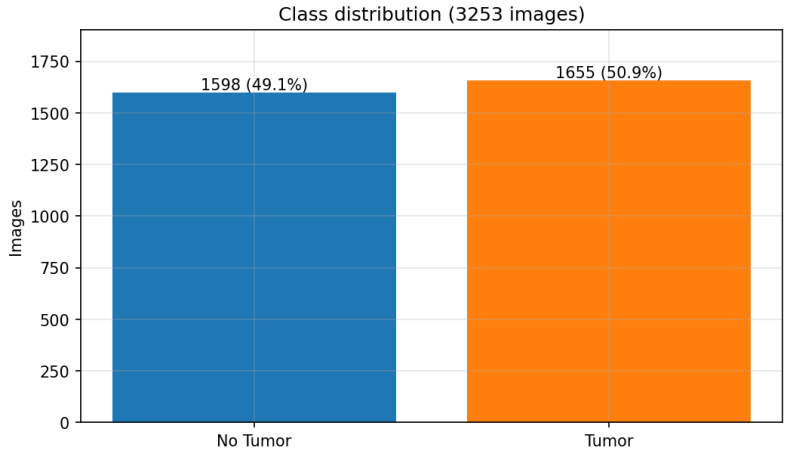
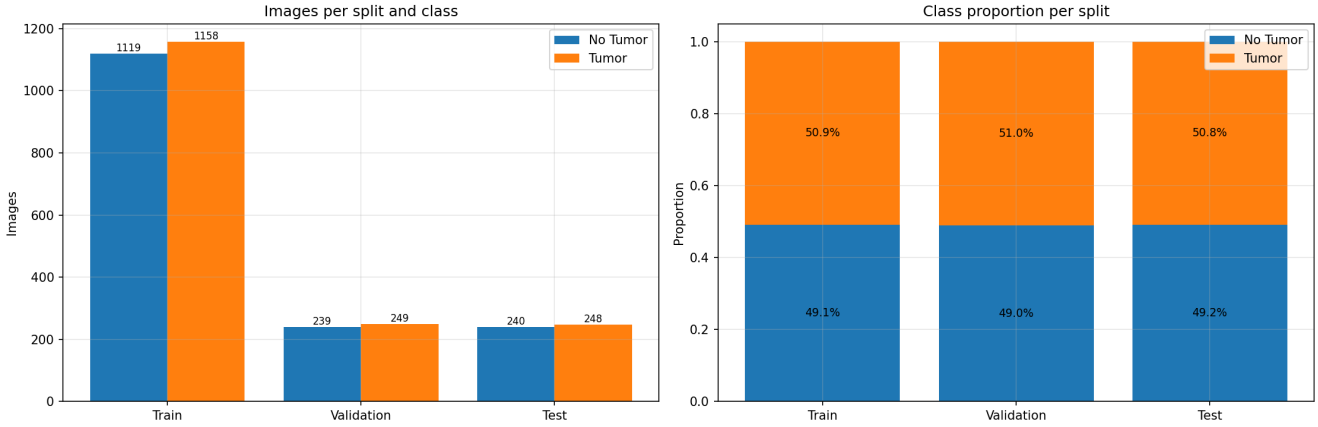
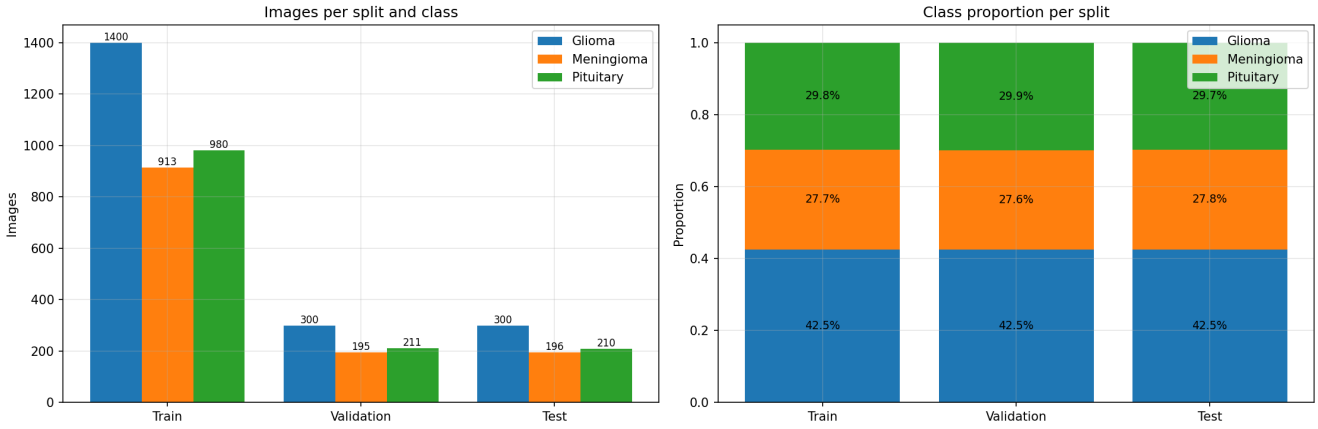


Figure S3: Class distribution of the combined binary detection dataset, comprising 1,598 tumour-negative and 1,655 tumour-positive images.



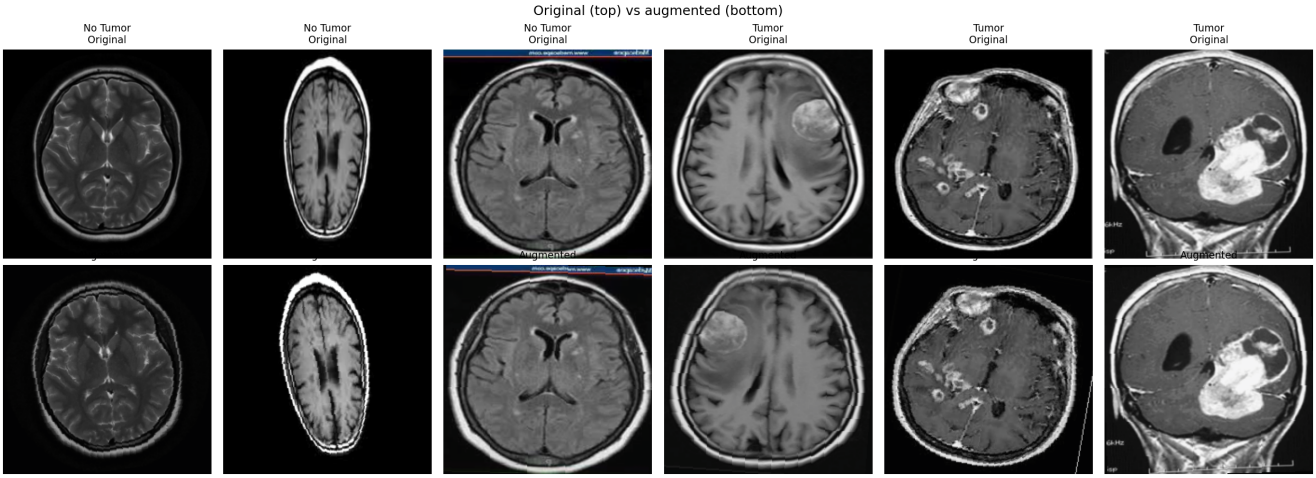
(a) Binary detection.



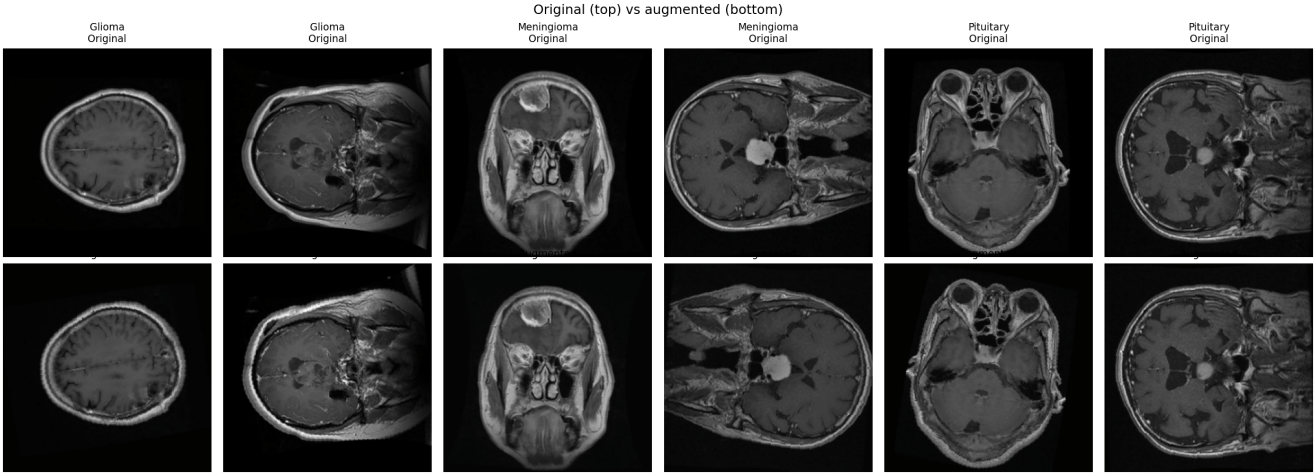
(b) Multiclass classification, after deduplication.

Figure S4: Stratified training, validation, and test split distributions for both tasks, in the proportions 70%, 15%, and 15%.

S3. Preprocessing and augmentation



(a) Binary detection task.



(b) Multiclass classification task.

Figure S6: Representative MRI images before and after the training-time augmentation pipeline, comprising random horizontal flipping, random rotation of up to ± 10 degrees, and brightness and contrast jitter of up to 20%.

S4. Per-class metrics

Tables S4 and S5 give the per-class precision, sensitivity, specificity, and F1 score of every model on the held-out test split. Support is the number of test images in each class.

Table S4: Per-class binary detection metrics on the held-out test split.

Model	Class	Support	Prec. (%)	Sens. (%)	Spec. (%)	F1 (%)
CoAtNet-0	No tumour	240	100.00	98.75	100.00	99.37
	Tumour	248	98.80	100.00	98.75	99.40
DenseNet121	No tumour	240	98.77	100.00	98.79	99.38
	Tumour	248	100.00	98.79	100.00	99.39
EfficientNetB0	No tumour	240	100.00	98.75	100.00	99.37
	Tumour	248	98.80	100.00	98.75	99.40
ResNet50	No tumour	240	98.77	100.00	98.79	99.38
	Tumour	248	100.00	98.79	100.00	99.39
InceptionV3	No tumour	240	97.95	99.58	97.98	98.76
	Tumour	248	99.59	97.98	99.58	98.78
Swin-Base	No tumour	240	93.65	98.33	93.55	95.93
	Tumour	248	98.31	93.55	98.33	95.87
ViT-B/16	No tumour	240	96.14	93.33	96.37	94.71
	Tumour	248	93.73	96.37	93.33	95.03
MobileNetV3-Large	No tumour	240	84.21	100.00	81.85	91.43
	Tumour	248	100.00	81.85	100.00	90.02
VGG16	No tumour	240	86.45	90.42	86.29	88.39
	Tumour	248	90.30	86.29	90.42	88.25

Table S5: Per-class multiclass classification metrics on the held-out test split.

Model	Class	Support	Prec. (%)	Sens. (%)	Spec. (%)	F1 (%)
Swin-Base	Glioma	300	99.00	99.33	99.26	99.17
	Meningioma	196	98.95	95.92	99.61	97.41
	Pituitary	210	97.67	100.00	98.99	98.82
CoAtNet-0	Glioma	300	98.03	99.67	98.52	98.84
	Meningioma	196	98.43	95.92	99.41	97.16
	Pituitary	210	99.05	99.05	99.60	99.05
InceptionV3	Glioma	300	98.99	98.33	99.26	98.66
	Meningioma	196	96.00	97.96	98.43	96.97
	Pituitary	210	99.52	98.57	99.80	99.04
EfficientNetB0	Glioma	300	98.66	98.00	99.01	98.33
	Meningioma	196	95.92	95.92	98.43	95.92
	Pituitary	210	97.64	98.57	98.99	98.10
ResNet50	Glioma	300	98.99	97.67	99.26	98.32
	Meningioma	196	94.53	96.94	97.84	95.72
	Pituitary	210	98.56	98.10	99.40	98.33
ViT-B/16	Glioma	300	91.28	90.67	93.60	90.97
	Meningioma	196	83.50	85.20	93.53	84.34
	Pituitary	210	96.15	95.24	98.39	95.69

S5. Training dynamics

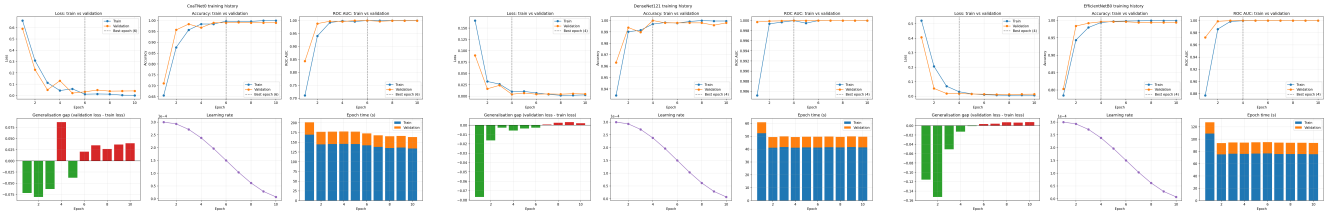
Table S6 reports the training and validation metrics at the best epoch for each model on the binary detection task, where the best epoch is the one giving the highest validation AUC. Table S7 reports the same quantities for the multiclass task at the checkpointed epoch, the one giving the highest validation macro one-versus-rest AUC. Because this metric saturates near 0.997, it distinguishes only weakly between late epochs, and for several models validation accuracy continued to rise after the checkpointed epoch; Swin-Base, for example, is checkpointed at epoch 7 with a validation accuracy of 97.31%, although its validation accuracy reaches 98.16% at epochs 9 and 10. These are validation-split figures; the corresponding test-split results are given in Tables 7 and 8 of the main text. Figures S7 and S8 show the training and validation curves of every model.

Table S6: Binary detection training and validation metrics at the best epoch, selected by validation AUC.

Model	Epoch	Train loss	Train acc.	Train AUC	Val. loss	Val. acc.	Val. AUC
CoAtNet-0	6	0.0114	0.9969	0.9999	0.0322	0.9898	0.9999
DenseNet121	4	0.0099	0.9969	1.0000	0.0042	1.0000	1.0000
EfficientNetB0	4	0.0307	0.9937	0.9996	0.0178	0.9959	0.9999
ResNet50	7	0.0042	0.9990	1.0000	0.0184	0.9939	0.9999
InceptionV3	3	0.0314	0.9922	0.9997	0.0106	0.9959	1.0000
Swin-Base	6	0.0953	0.9665	0.9938	0.0928	0.9693	0.9973
ViT-B/16	10	0.1649	0.9439	0.9837	0.1449	0.9406	0.9899
MobileNetV3-Large	10	0.0023	0.9990	1.0000	0.3995	0.8811	0.9993
VGG16	10	0.2590	0.8872	0.9603	0.2421	0.8975	0.9687

Table S7: Multiclass classification training and validation metrics at the checkpointed epoch, selected by validation macro one-versus-rest AUC.

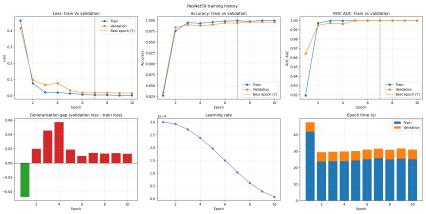
Model	Epoch	Train loss	Train acc.	Train AUC	Val. loss	Val. acc.	Val. AUC
Swin-Base	7	0.0333	0.9884	0.9996	0.0861	0.9731	0.9974
CoAtNet-0	9	0.0246	0.9928	0.9998	0.1019	0.9660	0.9975
InceptionV3	8	0.0132	0.9966	1.0000	0.1061	0.9745	0.9963
EfficientNetB0	9	0.0396	0.9893	0.9997	0.1069	0.9547	0.9959
ResNet50	9	0.0121	0.9954	1.0000	0.0939	0.9731	0.9981
ViT-B/16	10	0.3040	0.8808	0.9712	0.3419	0.8810	0.9689



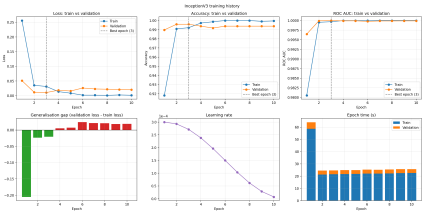
(a) CoAtNet-0.

(b) DenseNet121.

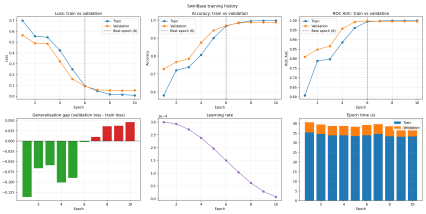
(c) EfficientNetB0.



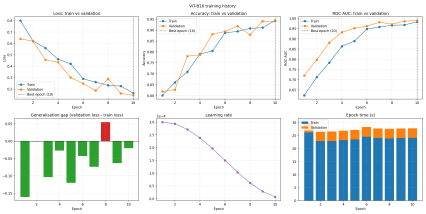
(d) ResNet50.



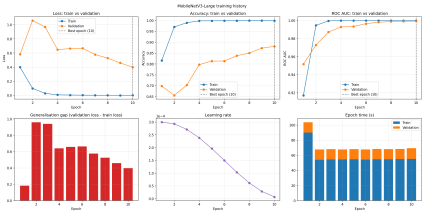
(e) InceptionV3.



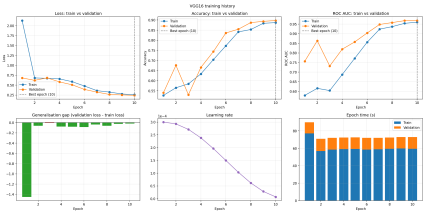
(f) Swin-Base.



(g) ViT-B/16.

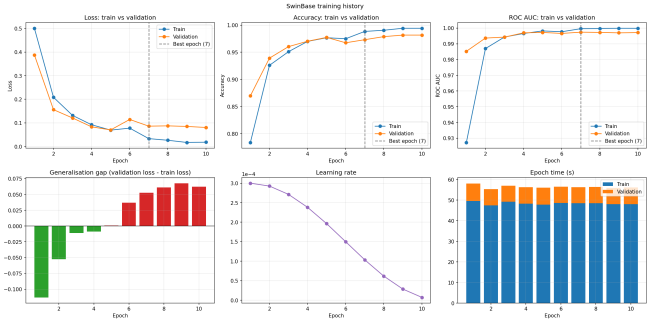


(h) MobileNetV3-Large.

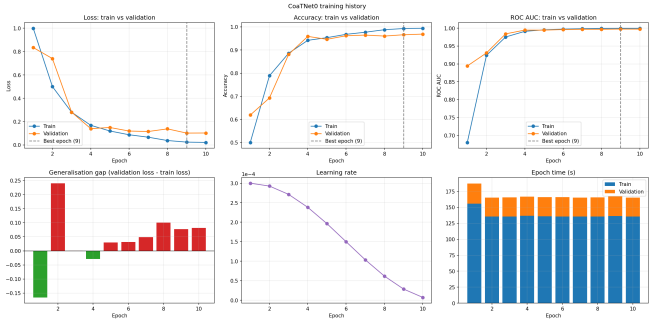


(i) VGG16.

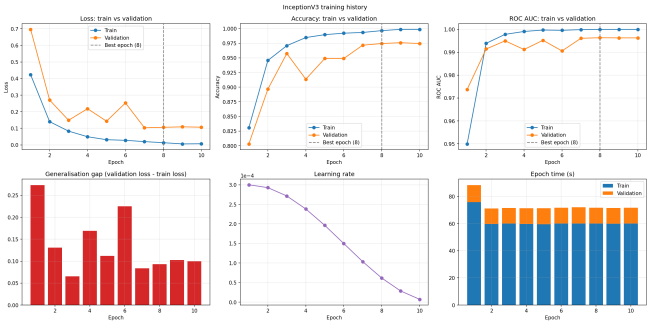
Figure S7: Training and validation loss, accuracy, and AUC over 10 epochs for all nine models on the binary detection task. The best validation AUC of EfficientNetB0, 0.9999, was reached at epoch 4.



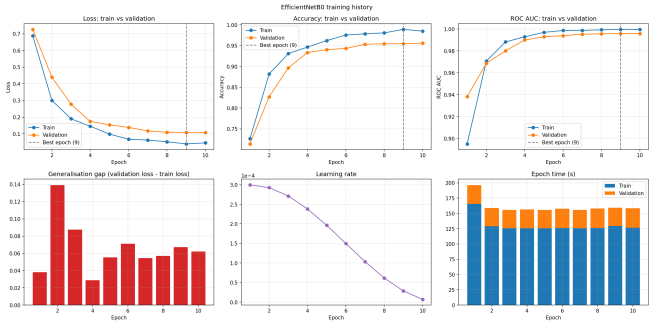
(a) Swin-Base.



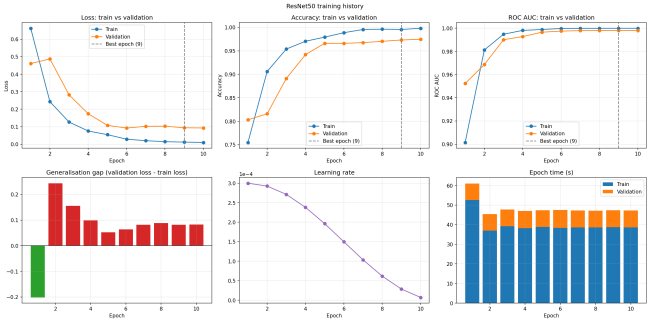
(b) CoAtNet-0.



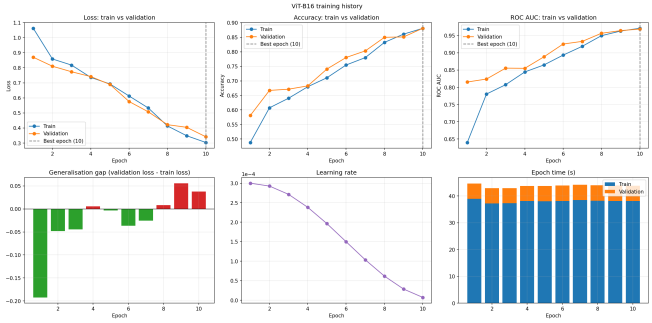
(c) InceptionV3.



(d) EfficientNetB0.



(e) ResNet50.



(f) ViT-B/16.

Figure S8: Training and validation loss, accuracy, and AUC over 10 epochs for all six models on the multiclass classification task. The checkpointed epoch of EfficientNetB0 is 9, selected on validation macro AUC, at which validation accuracy was 95.47%.

S6. Complete efficiency measurements

Tables S9 and S10 give the complete efficiency measurements for each task, of which the main-text efficiency table reports the paired subset. Training used dual NVIDIA Tesla T4 graphics processing units, except the binary InceptionV3 run, which executed on a single device and whose training time is therefore not comparable. Latency and throughput were measured on one device for every model under half-precision autocasting. Latency is the mean $\pm$ standard deviation of 100 timed single-image forward passes after 10 warm-up passes; throughput is measured at batch size 32 as the mean over 20 timed passes after 3 warm-up passes, for which per-pass timings were not retained. "Acc./M" is test accuracy in percentage points per million parameters. Rows are ordered by parameter count.

Table S9: Computational efficiency for the binary detection task. The InceptionV3 GFLOPs are computed at its native 299×299 input, and its training time is a single-GPU measurement.

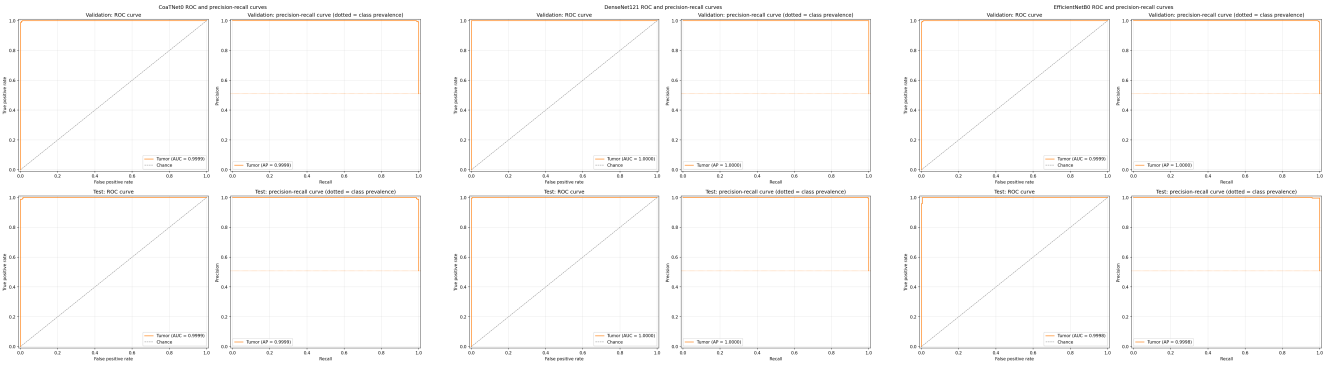
Model	Params (M)	GFLOPs	Batch	Train (min)	Latency (ms)	Thr. (img/s)	Acc./M
EfficientNetB0	4.01	0.77	256	16.3	10.8 ± 0.5	1364	24.79
MobileNetV3-Large	4.20	0.43	256	12.0	8.1 ± 0.6	2409	21.60
DenseNet121	6.95	5.67	64	8.5	19.4 ± 0.9	630	14.29
ResNet50	23.51	8.17	256	5.4	7.3 ± 0.5	771	4.23
InceptionV3	24.35	11.42	128	4.9	14.0 ± 1.0	514	4.06
CoAtNet-0	26.67	8.42	64	29.1	13.9 ± 0.7	520	3.73
ViT-B/16	85.80	22.54	128	4.7	8.8 ± 0.6	287	1.11
Swin-Base	86.74	30.86	64	6.6	29.0 ± 1.1	196	1.11
VGG16	134.26	30.93	256	12.5	8.2 ± 0.5	338	0.66

Table S10: Computational efficiency for the multiclass classification task. The InceptionV3 GFLOPs are computed at its native 299×299 input.

Model	Params (M)	GFLOPs	Batch	Train (min)	Latency (ms)	Thr. (img/s)	Acc./M
EfficientNetB0	4.01	0.77	256	26.9	13.9 ± 1.0	1471	24.33
ResNet50	23.51	8.17	256	8.1	7.8 ± 1.3	792	4.15
InceptionV3	25.12	11.42	128	12.2	14.2 ± 0.9	533	3.91
CoAtNet-0	26.67	8.42	128	28.0	13.2 ± 0.6	499	3.69
ViT-B/16	85.80	22.54	32	7.4	8.4 ± 0.6	290	1.05
Swin-Base	86.75	30.86	64	9.5	35.1 ± 2.6	202	1.14

S7. Evaluation figures

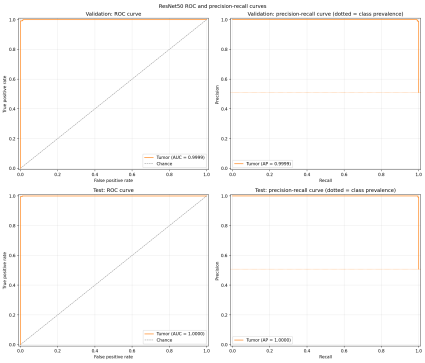
Figures S9 and S10 show receiver operating characteristic and precision–recall curves for every model on each task, and Figure S11 the reliability diagrams. Figures S12 and S13 show the per-class metric summaries and representative test-set predictions of EfficientNetB0, the released checkpoint, and Figure S14 the complete confusion-matrix panels whose test-split counts are shown in Figures 5–7 of the main text.



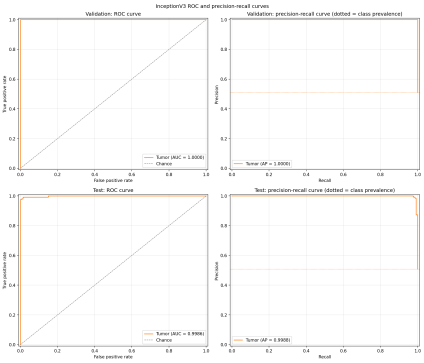
(a) CoAtNet-0.

(b) DenseNet121.

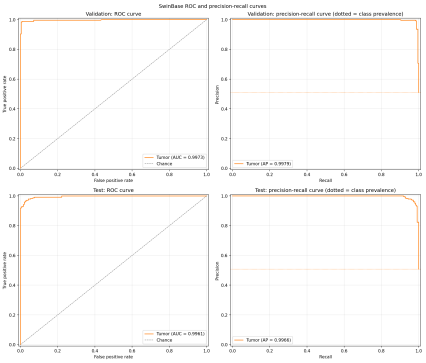
(c) EfficientNetB0.



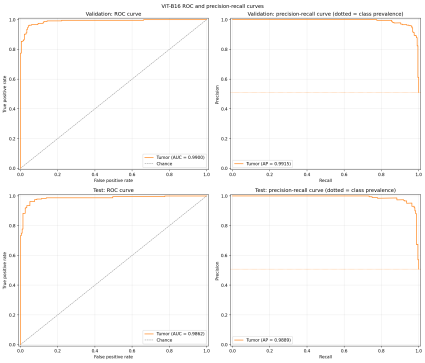
(d) ResNet50.



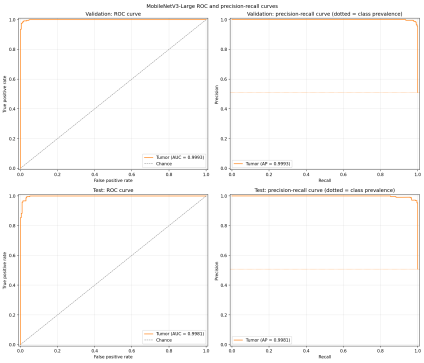
(e) InceptionV3.



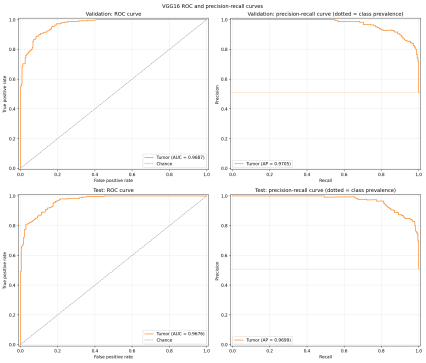
(f) Swin-Base.



(g) ViT-B/16.

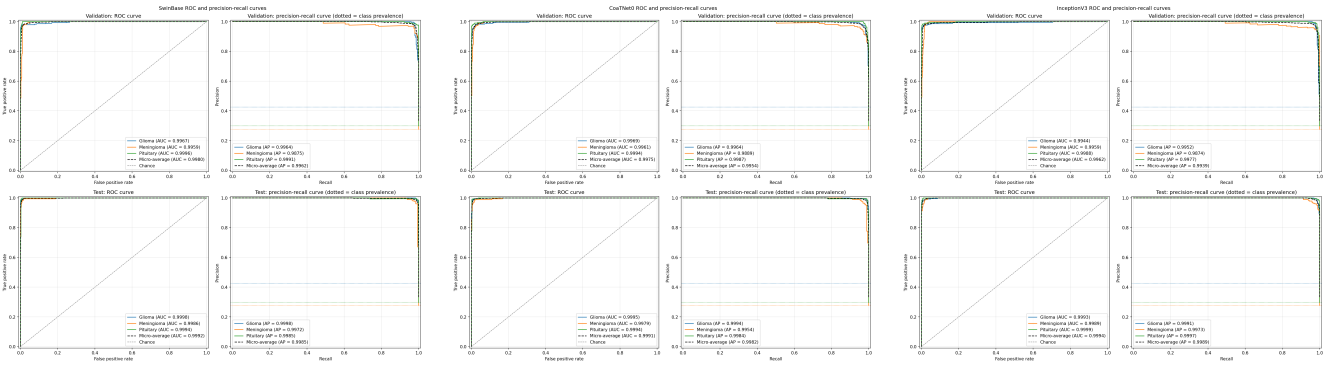


(h) MobileNetV3-Large.

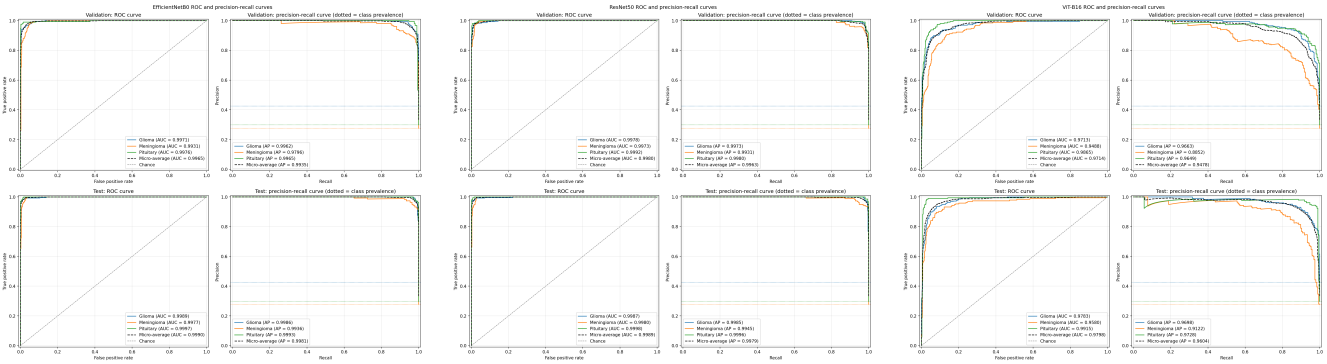


(i) VGG16.

Figure S9: Receiver operating characteristic and precision–recall curves for all nine models on the binary detection test split. EfficientNetB0 attains an AUC of 0.9998 and an average precision of 0.9998.

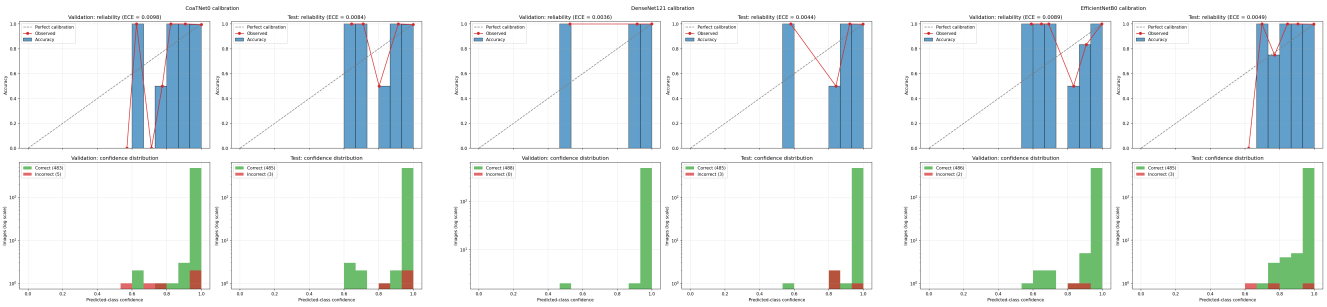


(a) Swin-Base. **(b)** CoAtNet-0. **(c)** InceptionV3.

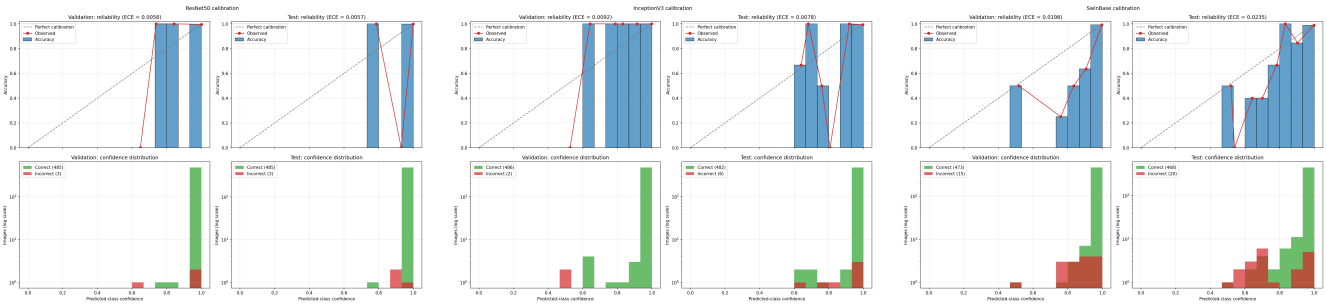


(d) EfficientNetB0. **(e)** ResNet50. **(f)** ViT-B/16.

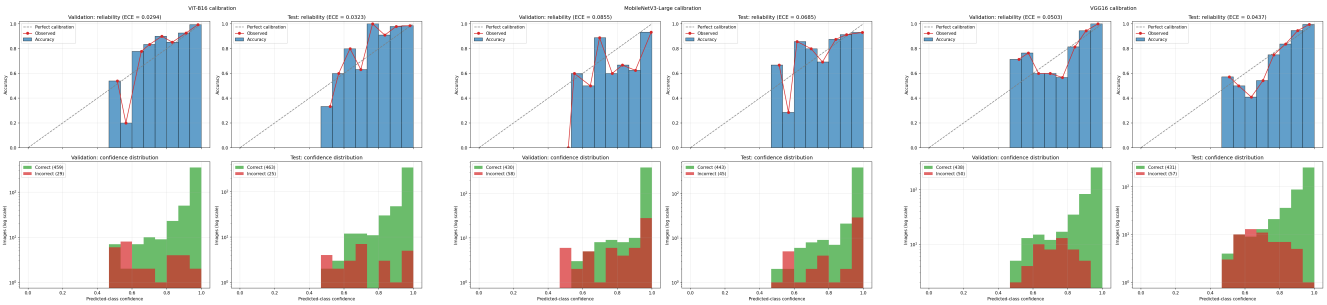
Figure S10: One-versus-rest receiver operating characteristic and precision–recall curves for all six models on the multiclass test split. Swin-Base and EfficientNetB0 attain macro-averaged AUCs of 0.9993 and 0.9988.



(a) Binary, CoAtNet-0. **(b)** Binary, DenseNet121. **(c)** Binary, EfficientNetB0.

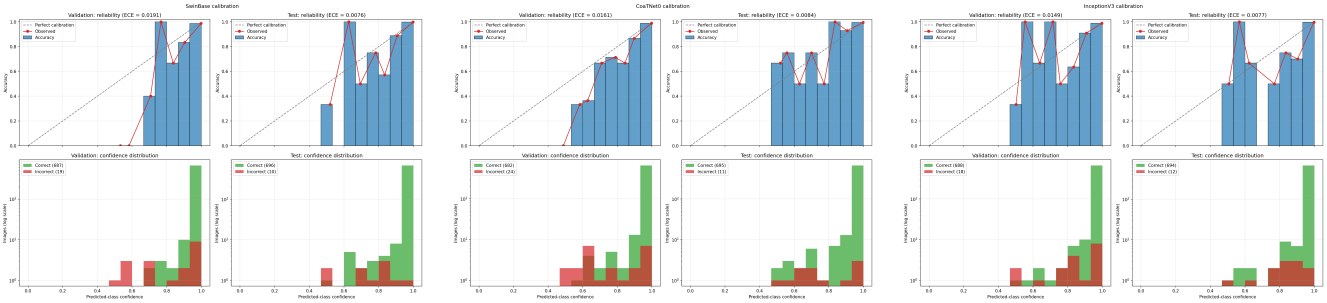


(d) Binary, ResNet50. **(e)** Binary, InceptionV3. **(f)** Binary, Swin-Base.



(g) Binary, ViT-B/16. **(h)** Binary, MobileNetV3-Large. **(i)** Binary, VGG16.

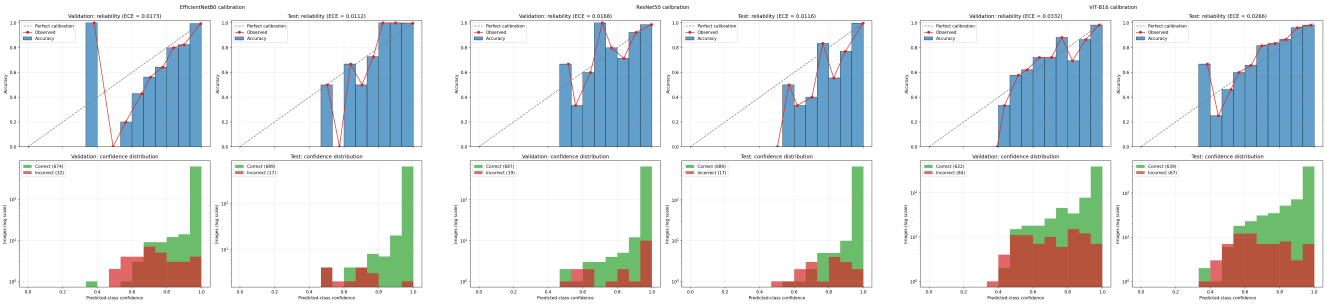
Figure S11: Reliability diagrams for all models on the binary detection test split (continued overleaf for the multiclass task). EfficientNetB0 attains an expected calibration error of 0.0049.



(j) Multiclass, Swin-Base.

(k) Multiclass, CoAtNet-0.

(l) Multiclass, InceptionV3.

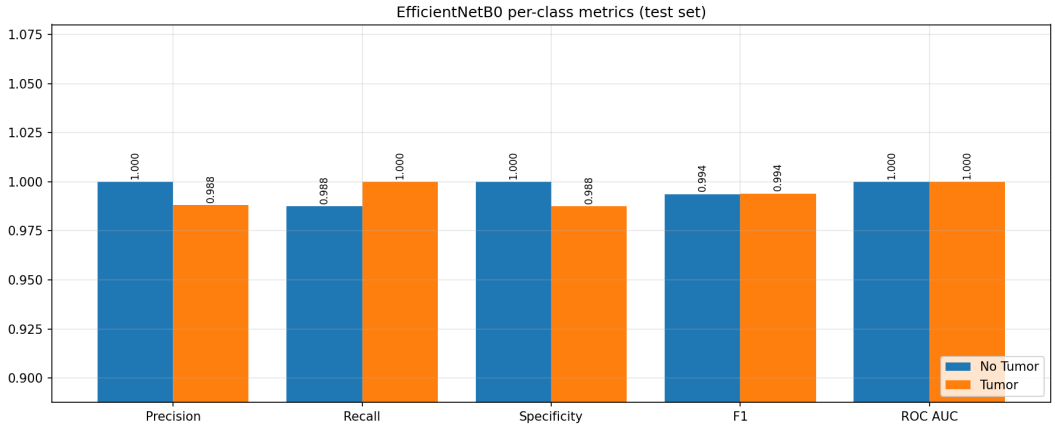


(m) Multiclass, EfficientNetB0.

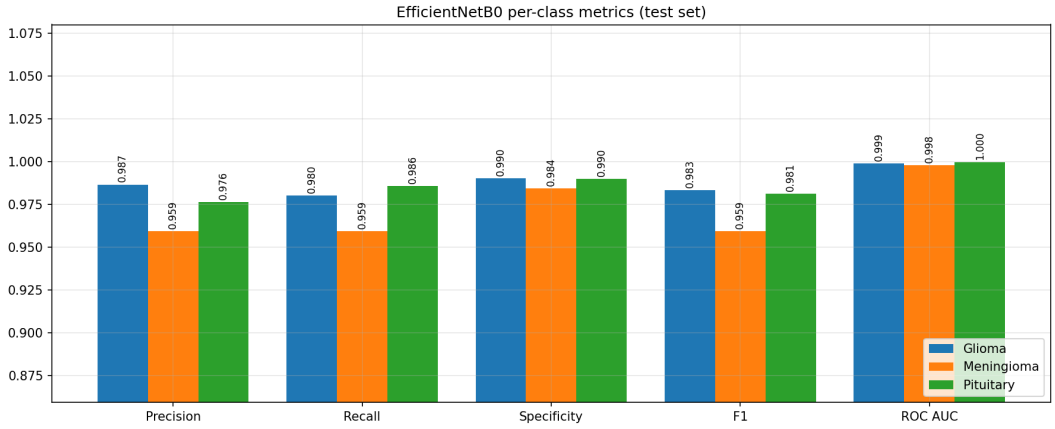
(n) Multiclass, ResNet50.

(o) Multiclass, ViT-B/16.

Figure S11: (continued) Reliability diagrams for all models on the multiclass test split. EfficientNetB0 attains an expected calibration error of 0.0112.

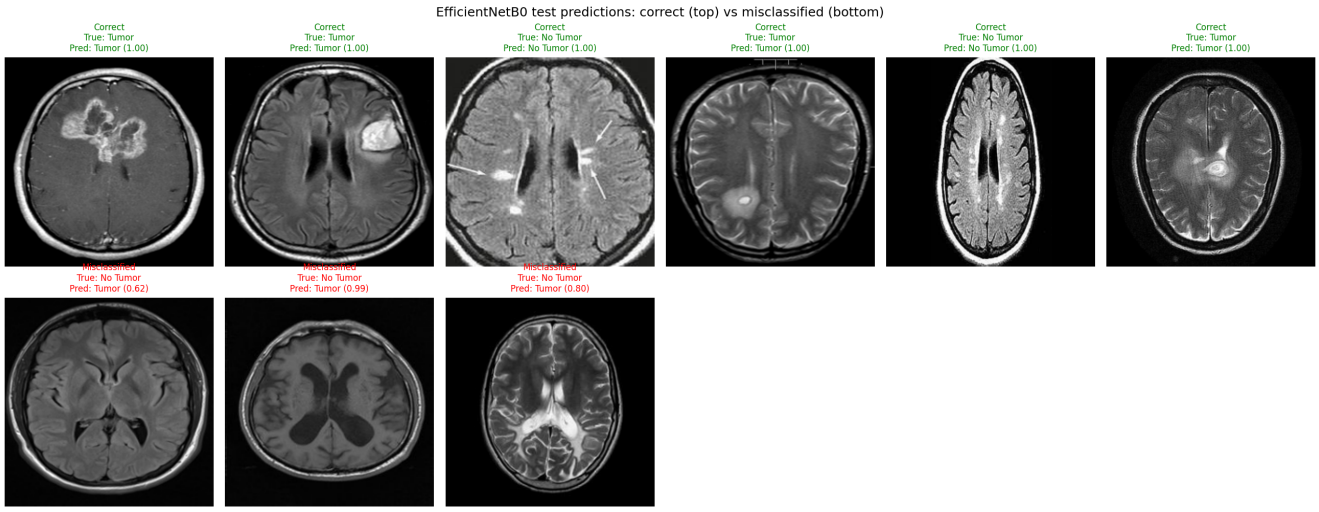


(a) Binary detection. The tumour class attains perfect recall, with the residual error confined to precision on that class.

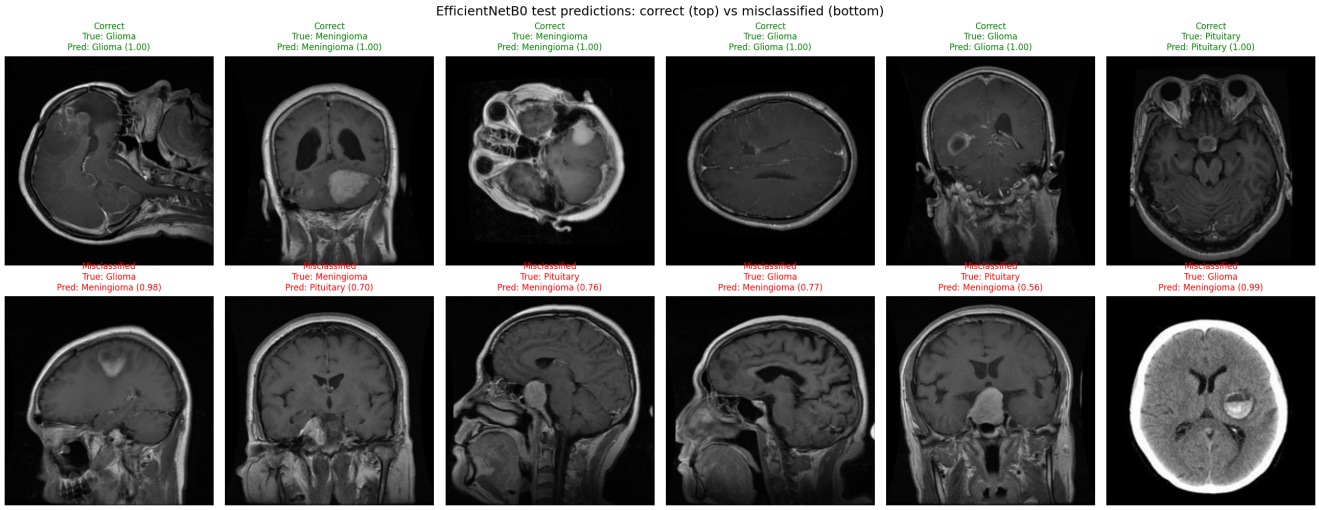


(b) Multiclass classification, showing meningioma as the weakest of the three classes.

Figure S12: Per-class precision, recall, and F1 scores for **EfficientNetB0** on the test split of each task.

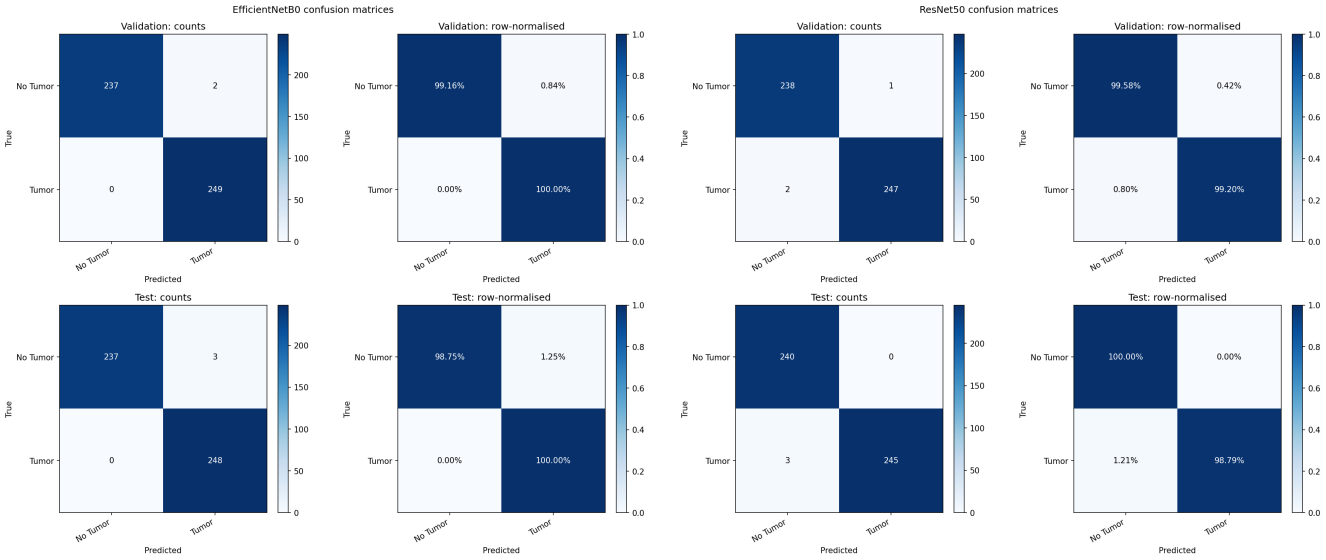


(a) Binary detection.



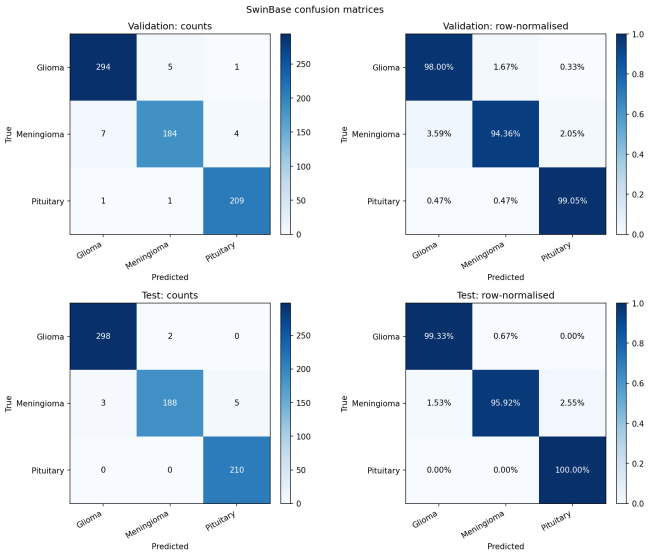
(b) Multiclass classification.

Figure S13: Representative test-set predictions of **EfficientNetB0**, with the predicted class and associated probability shown for each image.

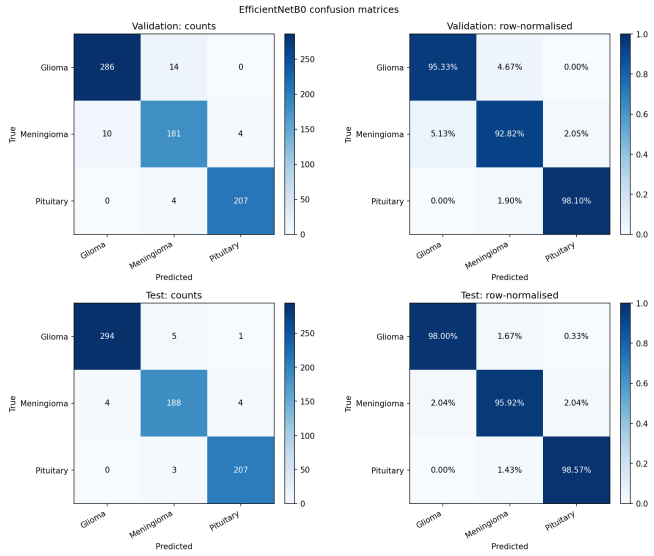


(a) EfficientNetB0, binary detection.

(b) ResNet50, binary detection.



(c) Swin-Base, multiclass classification.

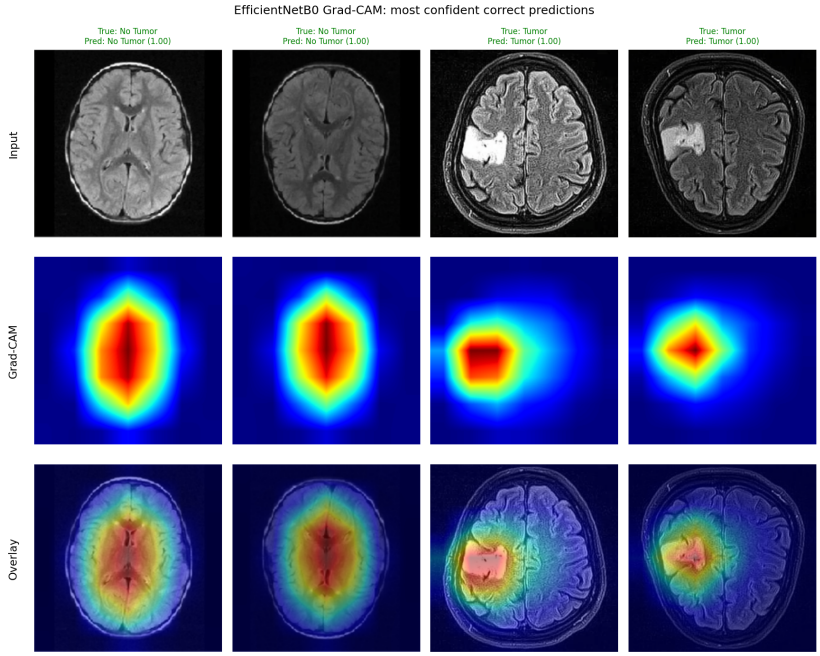


(d) EfficientNetB0, multiclass classification.

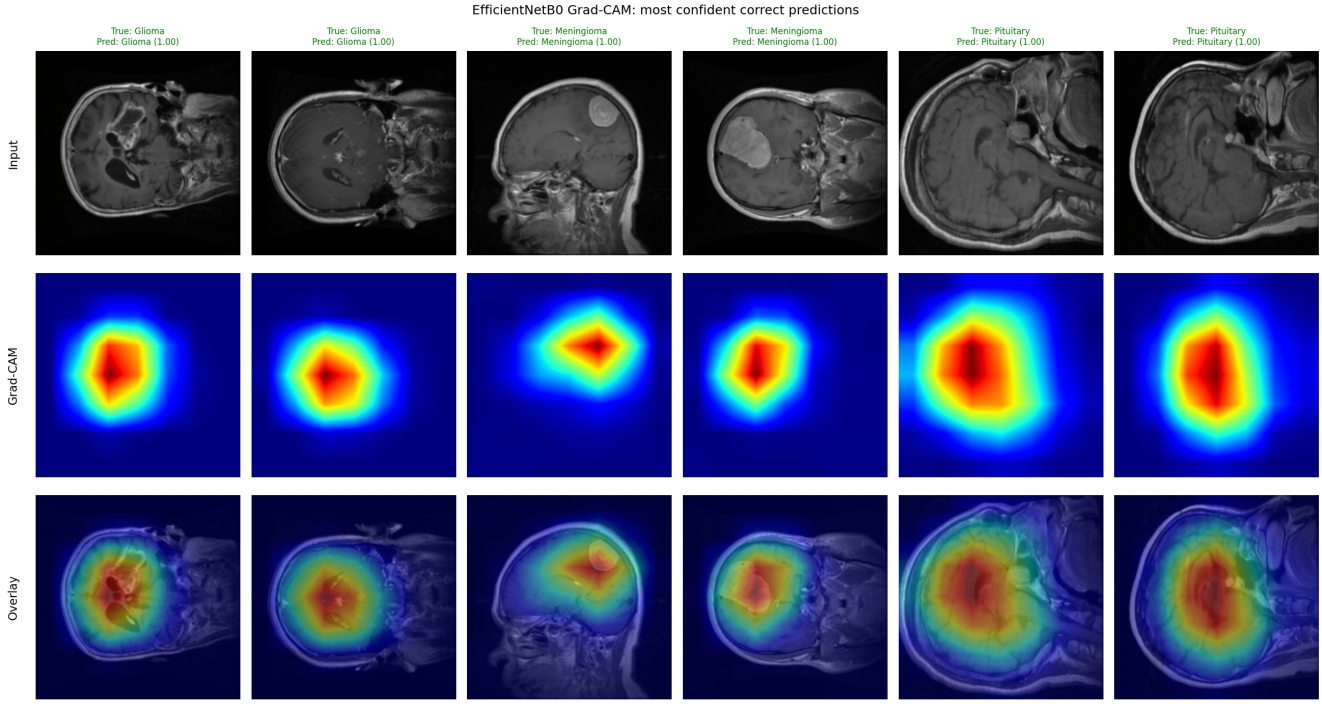
Figure S14: Complete confusion-matrix panels for the validation and held-out test splits, as counts and row-normalised, for the models shown in Figures 5–7 of the main text.

S8. Gradient-weighted class activation maps

Figure S15 gives the complete input, activation, and overlay panels for the most confident correct EfficientNetB0 predictions, whose overlay rows appear in Figure 8 of the main text. Figure S16 shows the most confident misclassified cases, up to six per model, ranked by the probability assigned to the wrong class; these are the failures in which the model was most assured, not representative ones.

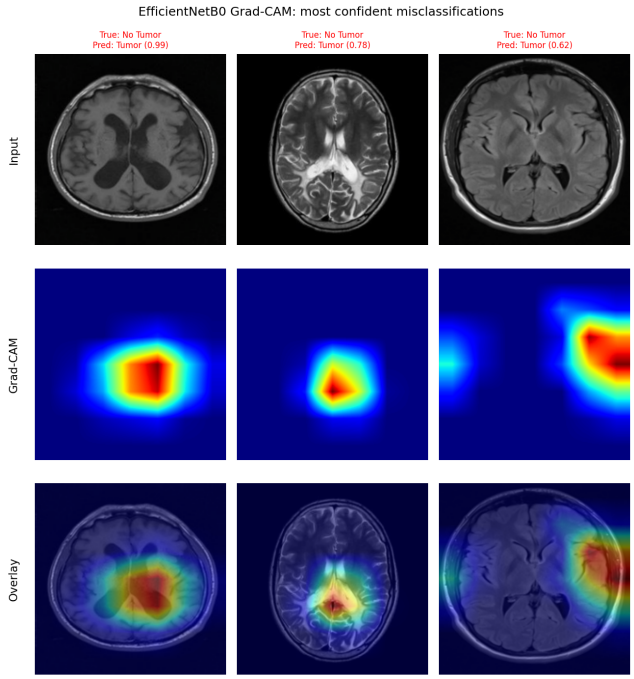


(a) Binary detection, showing the two most confident correct predictions per class.

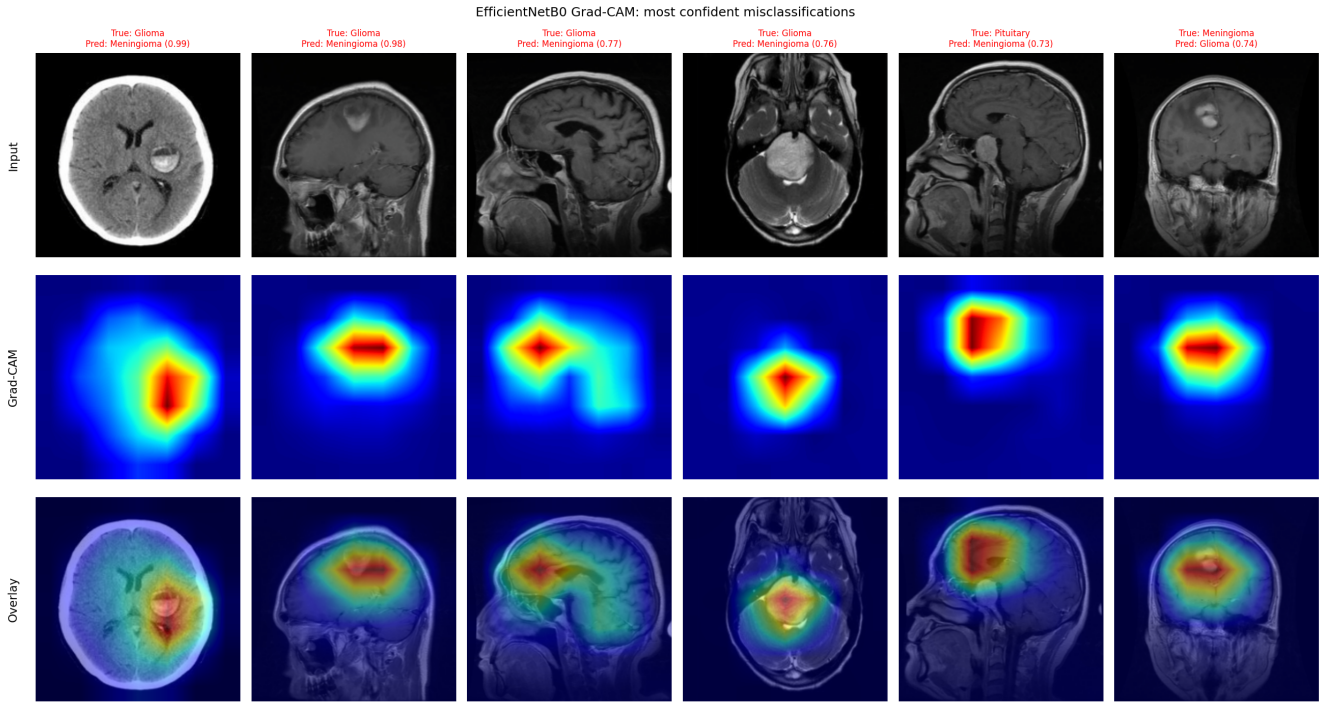


(b) Multiclass classification, showing the two most confident correct predictions for each of glioma, meningioma, and pituitary.

Figure S15: Gradient-weighted class activation maps for **EfficientNetB0** on correctly classified test cases.



(a) Binary detection. Only three test images were misclassified, so all are shown.



(b) Multiclass classification: the six most confident of seventeen errors.

Figure S16: Gradient-weighted class activation maps for **EfficientNetB0** on misclassified test cases, ranked by confidence in the wrong class.